WEEK 12 OF 15 · STAGE 3: ADVANCED METHODS

Why models disagree

Same question, different models, different answers

New R skill Reproducible analysis: folders, sourced scripts, Git basics

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	Week 9 Land, water and bioenergy	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	**Week 12 Why models disagree**	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Name the three sources of difference: structure, inputs and method
2. Explain how an intercomparison project makes model results comparable
3. Compare a GCAM run with AR6 scenarios after matching units and scope
4. Organise an analysis into sourced functions and save it with Git

? **Big question** If several respected models get the same carbon price and give different emissions, which one should we believe, and what does the spread tell us?

Same carbon price, three different futures

Three models get the same carbon price path. By 2050 one has cut CO2 by 60%, one by 40% and one by only 30%.



Is one of them wrong, or are they telling us something?

AN EVERYDAY PICTURE

Three cooks, one recipe name



Ask three cooks for "a chocolate cake" and you get three cakes. One has a different oven, one used different cocoa, one followed a different method. Hand them all the same recipe card and the differences that remain point to ovens and methods.

What it means for models

1. Different oven = different model structure
2. Different cocoa = different inputs
3. Different recipe steps = different method

KEY IDEA

Three reasons models give different answers

1

Structure

Which regions, sectors and technologies exist. A model with no land sector cannot trade bioenergy against food.

2

Inputs

Technology costs, resources, population, income. Your CheapSolar run changed one input and the whole mix moved.

3

Method

Simulation or optimisation, myopic or perfect foresight (week 11). The same data can give different plans.

HOW IT WORKS

How an intercomparison project works

1

Agree the task

Shared scenarios written down in a protocol

2

Each team runs it

Every model keeps its own structure and method

3

Report one way

Same variable names, same units, one template

4

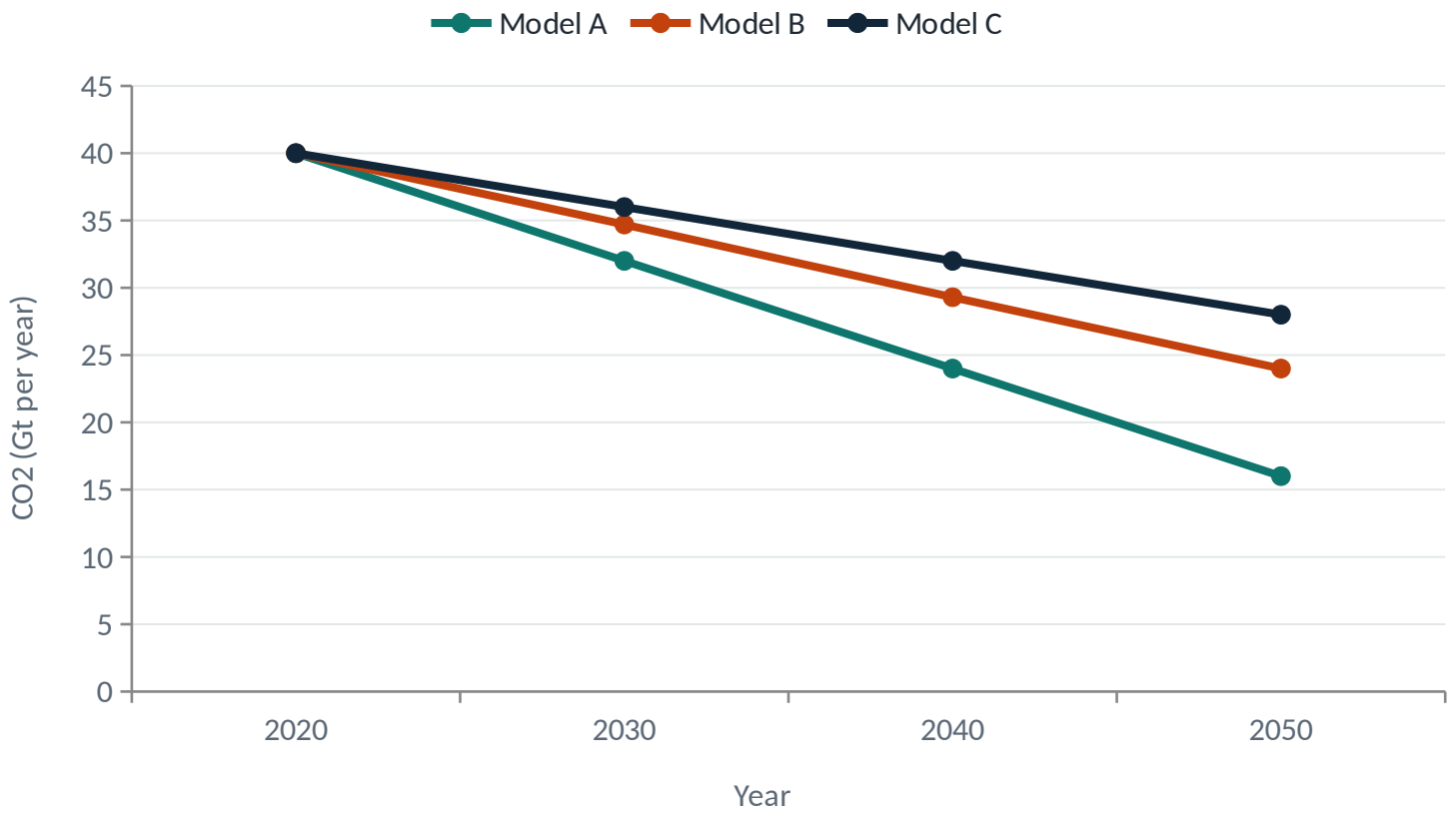
Compare, explain

Where do results agree, and why not?

The AR6 files you used in lab 6 share column names because teams agreed a reporting template

PICTURE THE NUMBERS

Same price path, three responses



Illustrative models with straight-line paths, not real results

Read it like this

1. All three start at 40 Gt in 2020
2. In 2050: A 16, B 24, C 28 Gt
3. The spread starts an explanation; it is not a probability

Worked example: a simple diagnostic indicator

Problem Three models start at 40 Gt in 2020. Under the same price, in 2050 A emits 16, B 24 and C 28 Gt. How much does each cut?

1. Cut = (2020 value − 2050 value) ÷ 2020 value
2. A: (40 − 16) ÷ 40 = 24 ÷ 40 = 60%
3. B: (40 − 24) ÷ 40 = 16 ÷ 40 = 40%
4. C: (40 − 28) ÷ 40 = 12 ÷ 40 = 30%
5. A responds 60 ÷ 30 = 2 times as strongly as C

Answer **Cuts of 60%, 40% and 30%: A responds twice as strongly as C**

Quick check

12 of 18 models show fast solar growth. What can you conclude?

A Fast solar growth is 67% likely

B Most models behave that way under these assumptions

C The other 6 models are wrong

D Solar growth is certain

Quick check: answer

12 of 18 models show fast solar growth. What can you conclude?

A. Fast solar growth is 67% likely

B. Most models behave that way under these assumptions ✓

C. The other 6 models are wrong

D. Solar growth is certain

Worked example: match the units before comparing

Problem GCAM reports a world total of 10,000 MtC (million tonnes of carbon). AR6 reports GtCO2. Convert.

1. A CO2 molecule weighs 44; its carbon atom weighs 12
2. So 1 tonne of carbon = 44 ÷ 12 = 3.667 tonnes of CO2
3. 10,000 MtC × 44 ÷ 12 = 36,667 MtCO2
4. 36,667 ÷ 1,000 = 36.7 GtCO2

Answer

10,000 MtC = 36.7 GtCO2

KEY POINTS

What one big comparison found: EMF 27

- 18 energy-economy and integrated models ran the same technology and climate scenarios
- They agreed on efficiency, electrification and fast clean-up of power
- They found transport and other non-electric uses hardest to clean up
- They found bioenergy and carbon capture especially valuable
- Agreement and disagreement are both findings

COMPARE

From a messy script to a reproducible analysis

Works on my laptop today

- One long script with copied code blocks
- Relies on objects left in the console
- Files named final2_really_final.R

Anyone can rerun it

- Functions live in the R folder and are sourced
- Runs in a fresh session (Session, Restart R)
- Every change saved as a Git commit with a message

WATCH OUT

Common mix-ups

Not quite	Better
Most models show X, so X is likely	Models share data and code. A spread describes the tools, not the odds.
Compare GCAM's MtC directly with AR6's MtCO2	Multiply by 44/12 first, and compare energy-and-industry CO2 with the same scope.
A model that disagrees with the others is broken	It may have a different structure, input or method. Find out which before judging.

NEW WORDS THIS WEEK

Words to add to the class glossary

Model intercomparison A project where many modelling teams run the same agreed scenarios and report results in one shared format, so differences can be explained.

Model structure How a model is built: which regions, sectors and technologies exist and how they connect. It is fixed in the code.

Diagnostic indicator A standard number recorded for every model under the same test, such as how much emissions fall under one carbon price path.

Reproducible analysis An analysis that anyone can rerun from the folder of files and scripts and get the same tables and figures.

Git A tool that saves snapshots (commits) of your files with a short message, so you can see and undo every change.

HANDS-ON LAB

Lab 12: your GCAM run among AR6 scenarios

Turn repeated code into two functions, place your CarbonPrice run among AR6 scenarios with a similar carbon price, and save the work with Git.

Script: labs/lab12.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Turn the lab 6 reshape into a function

to_long gives any AR6 variable in long form. Fill GAP 1 with the function's argument, then move the block to R/ar6_helpers.R.

```
to_long <- function(ar6, variable) {
  ar6 |>
    filter(Variable == ___) |>   # GAP 1
    pivot_longer(matches("^[0-9]{4}$"),
      names_to = "year", values_to = "value") |>
    mutate(year = as.integer(year)) |>
    drop_na(value)
}
```

Fill each ___ gap before running that line. Stuck? The solution file has the complete version.

Read GCAM's CO2 in GtCO2

get_co2 sums all regions and converts million tonnes of carbon to GtCO2 with 44/12 and a division by 1,000.

```
get_co2 <- function(prj, scen) {
  getQuery(prj, "CO2 emissions by region",
           scenarios = scen) |>
    group_by(scenario, year) |>
    summarise(gt = sum(value) * 44 / 12 / 1000,
              .groups = "drop")
}
prj  <- loadProject("output/gcam_course.dat")
gcam <- get_co2(prj, "CarbonPrice")
```

LAB · STEP 3 OF 4

Pick AR6 runs with a similar price

Choose a band around your 2050 carbon price, for example half to double. Fill GAPS 2 and 3.

```
similar <- to_long(ar6, "Price|Carbon") |>
  filter(year == 2050,
         value > ___, value < ___)  # GAPS 2, 3
nrow(similar)          # aim for 20 to 100
ar6_co2 <- to_long(ar6, co2_var) |>
  semi_join(similar, by = c("Model", "Scenario")) |>
  mutate(run = paste(Model, Scenario))
```

⌨ *Fill each ___ gap before running that line. Stuck? The solution file has the complete version.*

LAB · STEP 4 OF 4

Save your work with Git

Type these in the Terminal pane, not the Console. Copy gitignore_lab12.txt to .gitignore first.

```
git init
git status
git add .gitignore R labs
git commit -m "write your own message here"
git log --oneline
```

Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.

Did it work?

You should see

- nrow(similar) gives 20 to 100 scenarios
- Your orange GCAM line sits among the grey AR6 lines, not at a quarter of them
- git log --oneline shows your commit

Worksheet questions

1. Is your 2050 CO2 inside the AR6 range for a similar carbon price? Give the numbers.
2. Give two reasons from structure, inputs or method why your GCAM run could differ from the AR6 median.
3. Which files did you leave out of Git, and why?

RECAP

Take-home messages

1. Models differ because of structure, inputs and method
2. Intercomparison projects fix inputs and a reporting format so the spread can be explained
3. A spread across models describes the tools, not the probability of an outcome
4. Match units (× 44/12) and scope before comparing GCAM with AR6
5. Functions in sourced files plus Git commits make an analysis reproducible

Next: week 13

Week 13 runs GCAM many times on purpose, and uses a decision tree to discover which assumptions lead to high emissions.

Practise

Interactive course, week 12: try the disagree explorer and the 5-question quiz.

Readings: Keppo et al. (2021), Exploring the possibility space, Environmental Research Letters; IPCC AR6 Working Group III, Annex III; GCAM v8.2 documentation: Emissions outputs, Community guidelines for articles using GCAM